Topic 3: Physical Layer - Media, Signals, Multiplexing & Switching

Course outline weeks 1-2: physical layer transmission media, switching methods, ISDN, terminal handling. Switching appears in 4 of 5 papers.

How often this topic is examined

Question asked	Paper(s)	Marks
Circuit vs packet (vs message) switching; examples; latency & jitter	2023 Q5b [10], ISA2024 Q2d [8], SE2024 Q2c [4], 2025 Q3a [9]	4-10
Types of transmission media (list / explain / diagram for enterprise)	2023 Q1e [4], 2025 Q2b.i [4], SE2024 Q4c [4]	4
Bounded vs unbounded media	SE2024 Q1b.i	2
How fibre optic transmits data (diagram) + why fibre beats copper	2023 Q3a [5], Q3b [6]	11
Multiplexing: concept & importance; components of a multiplexer; FDM vs TDM	2025 Q2b.ii [4], SE2024 Q1b [4], Q2a [5]	4-5
Transmission impairments (attenuation, distortion, noise)	SE2024 Q1d	3
Analogue vs digital signals	SE2024 Q3c.ii	2
Modulation vs encoding with examples	SE2024 Q5b	6
ADSL: why asymmetric; data + voice on the same line	ISA2024 Q5b	7
GSM vs CDMA	SE2024 Q1b.iii	3

1. Switching

> **EXAM QUESTION** 2025 Q3a **[9]**
> Explain the concept of switching in networks. Differentiate between message switching, packet switching, and circuit switching.

Switching is the technique of directing data from a source to a destination through intermediate nodes (switches/routers) when there is no direct link between every pair of devices. It avoids the cost of a full mesh.

Made by King Tine, the AI Engineer

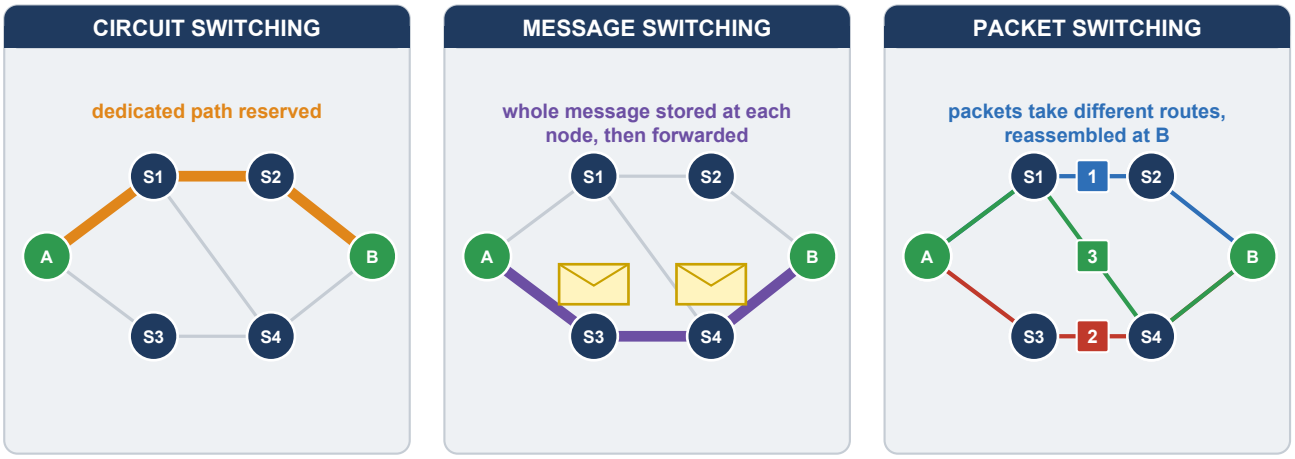


Figure: circuit vs message vs packet switching

Feature	Circuit switching	Message switching	Packet switching
Path	Dedicated physical path set up before data transfer and held for the whole call	No dedicated path; whole message stored at each node then forwarded	No dedicated path; message split into small packets, each routed independently (datagram) or along a virtual circuit
Phases	Setup -> data transfer -> teardown	Store-and-forward	Store-and-forward of packets
Resource use	Bandwidth reserved even when idle - wasteful	Better use of links, but nodes need large storage	Very efficient - links shared by many users (statistical multiplexing)
Delay	Setup delay, then constant low delay; no jitter	High delay (whole message waits at every node)	Variable delay (queuing) -> jitter; packets may arrive out of order
Failure	Link failure drops the call	Message can be rerouted	Packets rerouted around failure - robust
Charging	By time/distance	By message	By volume of data
Example	Traditional telephone (PSTN), ISDN	Telegraph, early email systems	Internet (IP), X.25, Frame Relay

EXAM QUESTION 2023 Q5b [4+4+2]; ISA2024 Q2d [4+4] **[10]**
i) Explain the main difference between a circuit-switched network and a packet-switched network. ii) Give two examples of packet-switched protocols and two examples of circuit-switched protocols. iii) Indicate which one would have higher latency and jitter and explain why.

i) In a **circuit-switched** network a dedicated end-to-end path with fixed bandwidth is established before communication and reserved for the whole session. In a **packet-switched** network no path is reserved: data is broken into packets that share links with other traffic and are forwarded independently (store-and-forward), possibly along different routes.

ii) Packet-switched protocols: **IP (IPv4/IPv6)**, **X.25**, **Frame Relay**, ATM, Ethernet, MPLS. Circuit-switched: **ISDN**, **PSTN/POTS signalling (SS7)**, **GSM (2G voice)**, T1/E1 (TDM), SONET/SDH.

iii) Packet switching has higher (variable) latency and jitter, because packets wait in router queues of different lengths, can take different routes and each node must store-and-forward them; so the delay changes from packet to packet. Circuit switching has a fixed dedicated path, so after setup the delay is constant (low jitter).

Circuit switching (SE2024 Q2c.i, 2 marks): dedicated path established for the duration of communication, e.g. telephone call. **Packet switching (Q2c.ii):** data split into packets with headers that travel independently and are reassembled at destination, e.g. the Internet.

2. Transmission media

> **EXAM QUESTION** 2023 Q1e [4], 2025 Q2b.i [4], SE2024 Q1b.i [2] **[2-4]**
> List any four types of transmission media. / Explain two types of transmission media. / Outline the difference between bounded and unbounded transmission media.

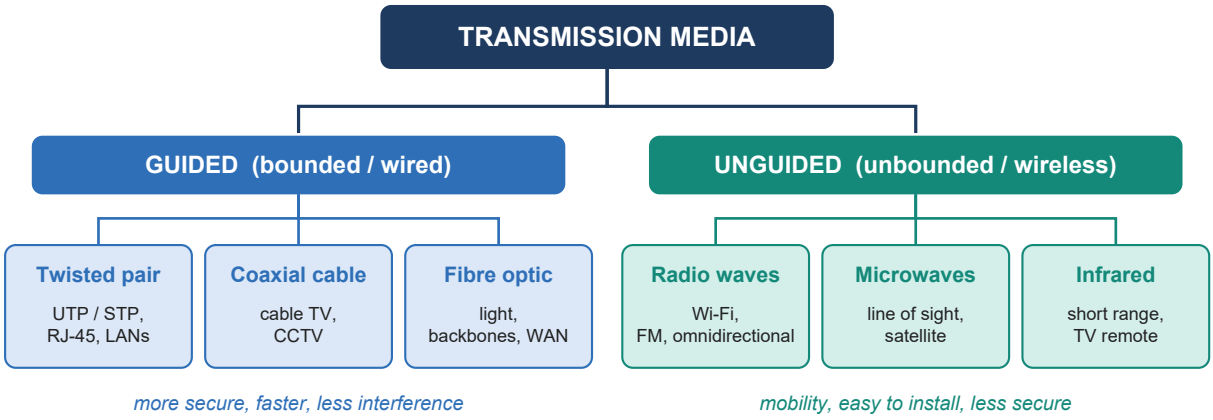


Figure: classification of transmission media

Bounded (guided)	Unbounded (unguided)
Signal travels along a **physical conductor** (copper/glass)	Signal travels through **air/space/vacuum** as electromagnetic waves
E.g. twisted pair, coaxial, fibre	E.g. radio, microwave, infrared, satellite
More secure, less interference, higher speed over the cable	Mobility, easy to install over long distances/difficult terrain; less secure, affected by weather & interference

Medium	Description	Use / features
Twisted pair (UTP/STP)	Pairs of insulated copper wires twisted to cancel crosstalk/EMI; RJ-45. Cat5e/Cat6	LANs, telephone lines; cheap, easy; max ~100 m; susceptible to noise
Coaxial cable	Central copper core, insulator, braided metal shield, outer jacket	Cable TV, older Ethernet (10Base2/5), CCTV; better shielding than UTP
Fibre optic	Thin glass/plastic core carries **light pulses** by total internal reflection; single-mode or multi-mode	Backbones, WANs, undersea cables; very high bandwidth, long distance, immune to EMI; expensive, fragile
Radio / microwave / satellite	Electromagnetic waves through air	Wi-Fi, mobile networks, point-to-point microwave links, VSAT

Media used in enterprise networks (SE2024 Q4c): UTP Cat6 for horizontal cabling to desks; **fibre** for the backbone between floors/buildings and to the ISP; **wireless (Wi-Fi 802.11)** for mobile users; draw a building with fibre risers, UTP to PCs and an access point.

3. Fibre optic cable

EXAM QUESTION 2023 Q3a [5], Q3b [6] **[11]**
Using a diagram, explain how data is transmitted along a fibre optic cable. / Identify three physical characteristics of fibre optic cables that make them more suitable for high-speed digital data transmission over copper cables.

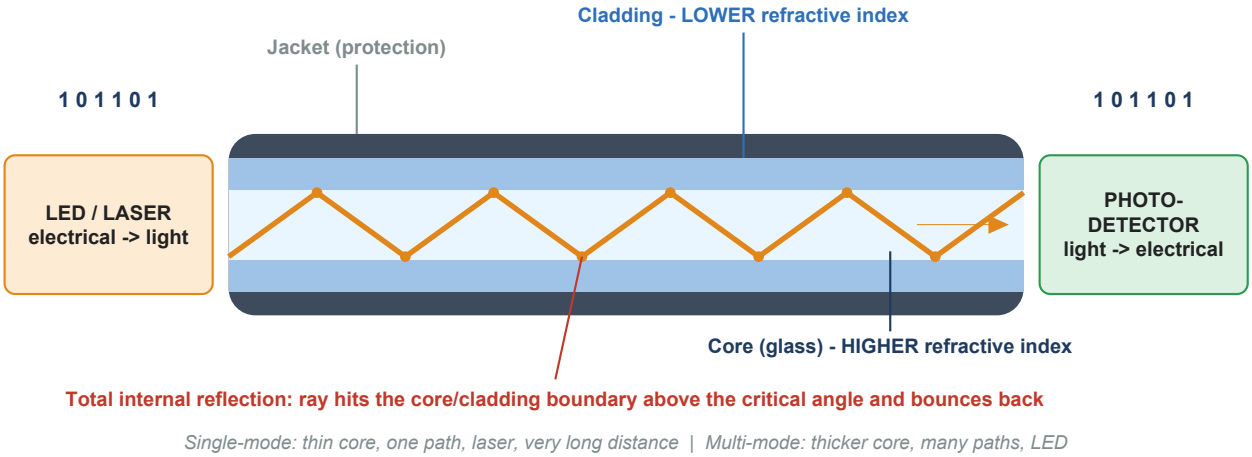


Figure: how light travels through a fibre optic cable

1. At the transmitter the electrical digital signal (1s and 0s) is converted into **light pulses** by an LED or laser diode (light ON = 1, OFF = 0).
2. The light enters the **core** (glass with a high refractive index) at an angle greater than the **critical angle**.
3. It is surrounded by **cladding** with a lower refractive index, so the light undergoes **total internal reflection** and bounces along the core with very little loss.
4. Buffer coating and outer jacket protect the fibre.
5. At the receiver a **photodetector (photodiode)** converts the light pulses back into electrical signals.
6. Types: **single-mode** (thin core, one light path, laser, very long distance) and **multi-mode** (thicker core, many paths, LED, shorter distances).

Why fibre is better than copper for high-speed data:

- **Very high bandwidth** - light has extremely high frequency, giving Gbps-Tbps data rates.
- **Low attenuation** - signals travel tens of km without repeaters (copper UTP ~100 m).
- **Immune to electromagnetic interference (EMI) and crosstalk** because it carries light, not electricity - fewer errors.
- **Thin, light, and secure** - hard to tap without detection; no electrical sparks/ground loops.

4. Signals: analogue vs digital; impairments

Analogue signal	Digital signal
Continuous wave with **infinite** values in a range	Discrete - **limited** number of values (e.g. 0 and 1)
Described by amplitude, frequency, phase	Described by bit rate & bit interval/level
E.g. human voice, analogue telephone, AM/FM radio	E.g. computer data, Ethernet signal
More affected by noise; noise is hard to remove	Can be regenerated by repeaters; more noise-immune

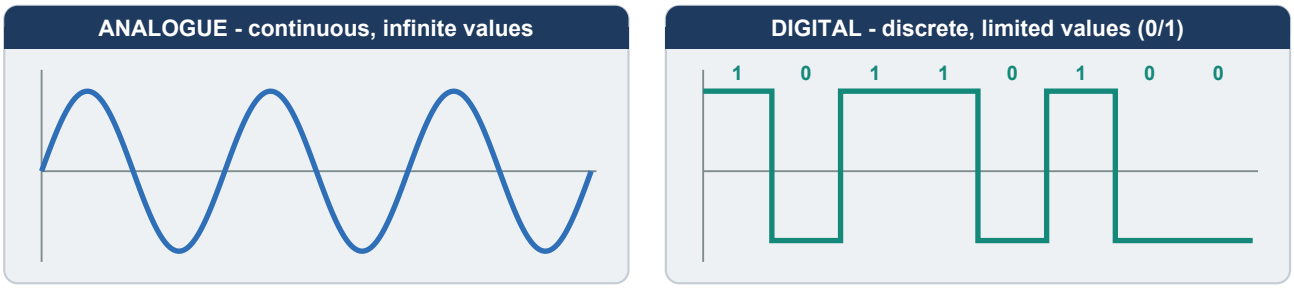


Figure: analogue vs digital signals

EXAM QUESTION SE2024 Q1d **[3]**
Signals pick up unwanted random energy; discuss three such transmission impairments.

- **Attenuation** - loss of signal energy/strength as it travels through the medium (resistance). Fixed with **amplifiers/repeaters**. Measured in decibels (dB).
- **Distortion** - the signal changes shape because its different frequency components travel at different speeds and arrive at different times (common in composite signals/fibre = dispersion).
- **Noise** - unwanted random energy added to the signal: **thermal** (random electron motion), **induced** (motors, appliances), **crosstalk** (one wire affecting another), **impulse** (spikes from lightning, power lines). Signal-to-noise ratio (SNR) measures quality.

Extra: **Shannon capacity** C = B log2(1 + SNR); **Nyquist** (noiseless) bit rate = 2 x B x log2(L). Bandwidth = range of frequencies a channel passes (Hz).

5. Modulation vs encoding

EXAM QUESTION SE2024 Q5b **[6]**
Distinguish between modulation techniques versus encoding. Give two examples of each.

Modulation	Encoding (line coding)
Converts digital (or analogue) data into an **analogue signal** by varying a **carrier wave**'s amplitude, frequency or phase	Converts digital data into a **digital signal** (pattern of voltage levels)
Used for long-distance / wireless / telephone lines (bandpass channels)	Used for short-distance baseband transmission (LAN cables)
Examples: **ASK** (amplitude shift keying), **FSK** (frequency), **PSK** (phase), **QAM**; analogue: AM, FM	Examples: **NRZ** (Non-Return-to-Zero), **Manchester** (Ethernet), Differential Manchester, AMI, RZ
Device: modem (modulator-demodulator)	Device: NIC / line coder

6. Multiplexing

EXAM QUESTION 2025 Q2b.ii [4], SE2024 Q1b [4] **[4]**
Describe the concept of multiplexing and its importance in data communication. / Describe the two components that make up a multiplexer.

Multiplexing is a technique that combines **multiple signals** from several sources into **one composite signal** so they can share a **single link/medium**. A **demultiplexer** at the other end separates them.

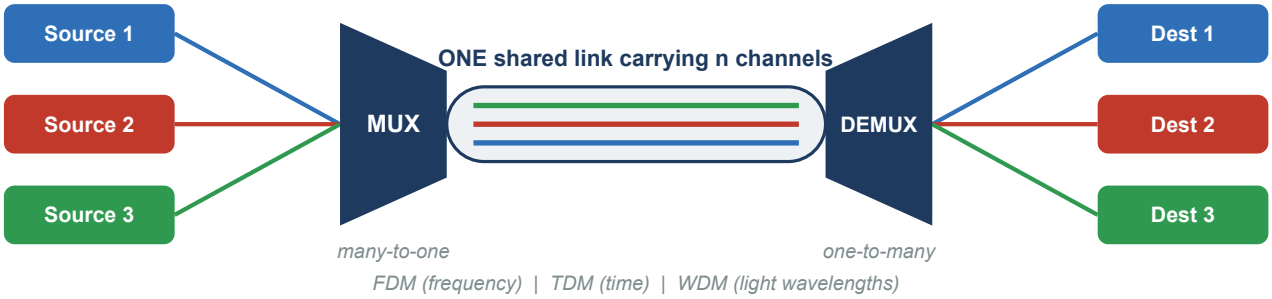


Figure: multiplexer and demultiplexer sharing one link

- **Two components:** (1) the **multiplexer (MUX)** - many-to-one: combines n input lines into one output stream; (2) the **demultiplexer (DEMUX)** - one-to-many: separates the stream and delivers each signal to its correct output line. (Between them is the shared link carrying n channels.)
- **Importance:** efficient use of expensive bandwidth; reduces cost (one cable instead of many); allows many users/calls to share a link (telephone trunks, cable TV, fibre backbones); simpler infrastructure.
- **Types: FDM**, **TDM** (synchronous & statistical), **WDM** (wavelength - for fibre), CDM.

EXAM QUESTION SE2024 Q2a **[5]**
Differentiate between Frequency Division Multiplexing (FDM) and Time Division Multiplexing (TDM).

FDM	TDM
Link bandwidth is divided into **frequency bands**; each signal gets its own band all the time	Link time is divided into **time slots**; each signal uses the whole bandwidth for its slot in turn
Mainly for **analogue** signals	Mainly for **digital** signals
Needs **guard bands** between channels to prevent overlap	Needs **synchronisation** (framing bits); guard time between slots
All signals transmitted **simultaneously**	Signals transmitted **one after the other** in rotation
Example: radio & TV broadcasting, cable TV, ADSL	Example: digital telephone T1/E1 lines, GSM (time slots)
Susceptible to crosstalk between bands	Wasted slots if a source has nothing to send (synchronous TDM) - fixed by **statistical TDM**



Figure: FDM vs TDM

7. ADSL

EXAM QUESTION ISA2024 Q5b **[2 + 5]**
i) Explain why ADSL is called asymmetric. ii) Describe how ADSL transmits both data and telephone calls over the same twisted pair without the two signals interfering.

i) It is **asymmetric** because the **download (downstream) rate is much higher than the upload (upstream) rate** (e.g. 8-24 Mbps down vs ~1 Mbps up) - more bandwidth is allocated downstream because home users download far more than they upload.

ii) ADSL uses **frequency division multiplexing** on the copper pair:

- **0-4 kHz**: ordinary voice telephone (POTS).
- **~25-138 kHz**: upstream data; **~138 kHz-1.1 MHz**: downstream data (DMT - discrete multitone - splits this into many 4 kHz sub-channels).
- Because voice and data occupy **different frequency bands**, they do not interfere.
- A **splitter / microfilter** at the home separates low frequencies (to the phone) from high frequencies (to the ADSL modem); at the exchange a **DSLAM** splits them again: voice to the telephone switch (PSTN), data to the ISP/Internet.

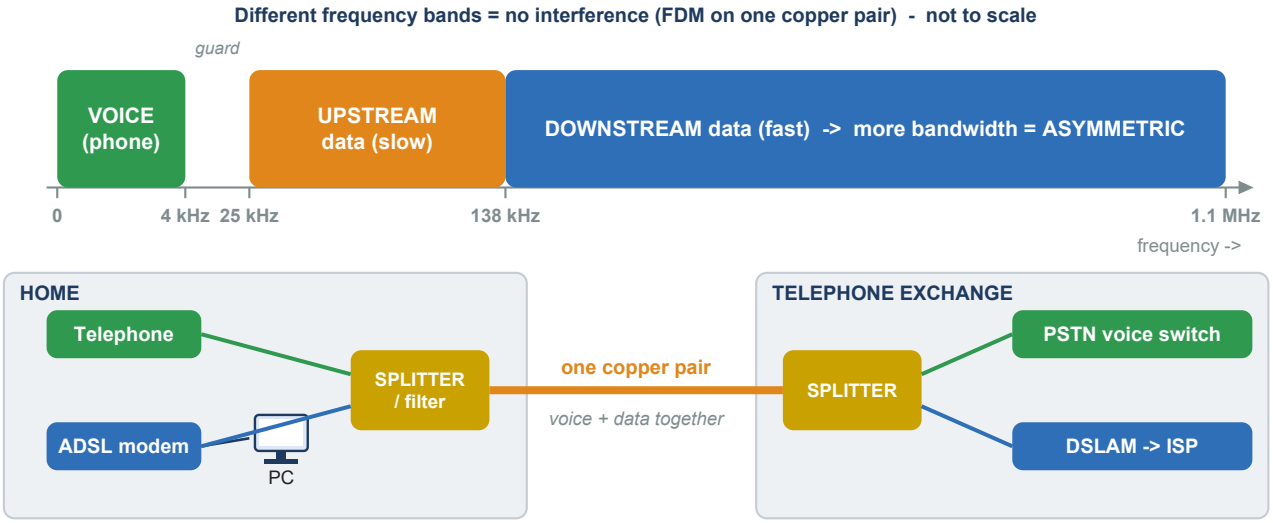


Figure: ADSL frequency bands and splitters

8. GSM vs CDMA (SE2024 Q1b.iii)

GSM (Global System for Mobile communication)	CDMA (Code Division Multiple Access)
Uses **TDMA + FDMA**: each frequency channel split into time slots (8 users per carrier)	All users share the **same frequency at the same time**; each separated by a unique **code** (spread spectrum)
Uses a **SIM card** - easy to switch phones/networks	Traditionally handset-bound (no SIM in early systems)
Worldwide standard (Europe, Africa - Econet, NetOne)	Used mainly in USA (Verizon, Sprint), parts of Asia
Hard handoff; simpler; can't use voice & data together on 2G	Soft handoff, better capacity and security; resistant to interference

Similarity: both are 2G/3G multiple-access digital cellular technologies; both superseded by 4G LTE (OFDMA).

9. ISDN & terminal handling (outline topics - may be asked)

- **ISDN (Integrated Services Digital Network)** - a circuit-switched digital telephone network that carries voice, data, fax and video over ordinary copper lines. **BRI** = 2B + D (2 x 64 kbps bearer + 16 kbps signalling = 144 kbps). **PRI** = 23B + D (T1) or 30B + D (E1).
- **Terminal handling** - connecting many dumb terminals to a host cheaply using **multiplexers, concentrators** and **polling/selection** (the host asks each terminal in turn whether it has data).

REMEMBER: Switching = 'how the path is found'; Multiplexing = 'how one link is shared'. Circuit = phone, Packet = Internet, Message = telegraph. FDM = frequency bands (radio), TDM = time slots (E1), WDM = colours of light (fibre).